

LEWEI XU

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SUMMARY

Honours student at UWA researching long document understanding, at the intersection of NLP and vision, with two publications in 2026 including one at a top-tier NLP conference (EMNLP). I specialise in NLP and deep learning but am also well versed in traditional machine learning for structured data and predictive modelling. I also have a strong data science background, working with data analysis, statistical modelling and visualisation. Systems and software development is another interest, now shifted more toward design and engineering as LLMs change what building software involves.

RESEARCH

TOPIC: MULTI-PAGE VISUALLY RICH DOCUMENT UNDERSTANDING

- Visually-rich documents (VRDs) such as scientific papers, financial reports and technical manuals are pervasive in real-world workflows and often span dozens to hundreds of pages. Automated document understanding allows organisations to save time, reduce manual review effort, and make faster, more informed decisions from large volumes of documents.
- MP-VRDU is the task of extracting or inferring information from long, visually-rich documents given their content, structure, and layout. In practice it is realised as document question answering, since summarisation, key information extraction, and structured output can all be framed as QA tasks.

PUBLICATIONS

- [Managing Evidence at Document Scale: A Survey of Multi-Page Visually Rich Document Understanding](#). Lewei Xu, Yihao Ding, Zihan Xu, Siwen Luo, Daochang Liu, Wei Liu. *Accepted in EMNLP Main Conference*. (arxiv link pending)
- [Locating Failure in Multi-Page Visually Rich Document Understanding: An Empirical Attribution](#). Lewei Xu, Yihao Ding, Zihan Xu, Daniel Yitian Su, Daochang Liu, Siwen Luo, Yifan Peng, Wei Liu. *Submitted to EACL, under review*.

RELEVANT EXPERIENCE

HARVEY ABORIGINAL CORPORATION

IT & Data Systems Intern

Harvey, WA
Nov 2025 – Dec 2025

- Designed and implemented new data and IT systems across the organisation
- Supported upgrades to software and hardware to improve reliability and efficiency
- Worked directly with staff to understand needs and provide practical system solutions
- Assisted with network and device setup, including configuring new hardware and maintaining data backup practices

AUSTRALIAN DEPARTMENT OF HEALTH

Data Scientist Intern

East Perth, WA
Jun 2025 – Nov 2025

- Contributed to WA health department's Synthetic Data Innovation Project
- Rewrote slow R libraries used in the evaluation pipeline of said project into Python
- Massively decreased run time of evaluation pipeline from ~30 minutes to less than 5 minutes
- Heavy use of data science libraries including Pandas, Numpy, Scikit-learn and Matplotlib
- Used Microsoft DevOps and Git to support version control and project workflows

UWA BUSINESS SCHOOL

Full Stack Developer

Nedlands, WA
Jun 2025 – Nov 2025

- Developed and deployed a complete software solution for an external client as part of an agile project team
- Built front-end components using HTML, CSS, and JavaScript and back-end services using Python (FastAPI) and SQLite
- Used Git for version control and collaborative development
- Deployed and maintained the application on AWS EC2, available at <https://g8-research-output-database.vercel.app/>

BANKSIA GROVE NEWSAGENCY

Manager

Banksia Grove, WA
May 2023 – Dec 2026

- Managed daily store operations, including staff rostering and shift coordination
- Responsible for cash handling, daily and weekly till balancing, and end-of-day reporting
- Monitored stock levels and placed orders to maintain appropriate inventory
- Delivered consistent, high-quality customer service in a front-facing role

EDUCATION

UNIVERSITY OF WESTERN AUSTRALIA

Bachelor of Advanced Computer Science (Honours)

Major in Artificial Intelligence; Minor in Data Science

Cumulative GPA: 6.56; WAM: 81

Relevant Coursework: Data Structures, Algorithms, Agile Web Development, Machine Learning, Deep Learning

Nedlands, WA

Jan 2023 – Nov 2026

DUNCRAIG SENIOR HIGH SCHOOL

Western Australia Certificate of Education

ATAR: 98.7

Subjects Completed: Mathematics Methods & Specialist, Physics, Chemistry, Computer Science, English

Duncraig, WA

Jan 2017 – Nov 2022

PERSONAL PROJECTS

LOGARIUM: MEDIA TRACKER

Feb 2024 (rebuilt 2026)

- Full-stack media tracker unifying ten media types into one library with consistent statuses, ratings, and cross-format statistics
- Concurrent metadata search across 15 external providers, with normalisation, deduplication, and graceful fallback on failure
- Service-oriented FastAPI/PostgreSQL backend with JWT multi-user auth and a personalised recommendation engine
- Live application available at <https://logarium.vercel.app/demo> (logged in demo account mirroring my personal account)

SEMANTIC NOVEL RECOMMENDER

Jun 2026

- A local, offline semantic search engine for Chinese web novels, embedding metadata into 1024-d vectors via BAAI/bge-m3 for cosine retrieval
- Hybrid ranking combining dense retrieval, a tag-overlap boost, and metadata filters, with an optional local LLM for query parsing and re-ranking
- Incremental indexing pipeline scaling to 100k+ novels, shipped with a FastAPI/React reader app
- Live application available at <https://novel-reader-rec.vercel.app/>

ACTIVITIES

UWA UNIMENTOR PROGRAM

UniMentor

Nedlands, WA

Feb 2026 – Jun 2026

- Selected to mentor first-year students, providing guidance on study planning, university systems, and transitioning into university life
- Acted as a point of contact for academic and general support, helping students navigate challenges and access resources

PROGRAMMING COMPETITION SOCIETY

Committee Member

Nedlands, WA

Jan 2023 – Jan 2025

- Participated in coding competitions, solving algorithmic and logic-based problems under time constraints
- Developed strong problem-solving, analytical thinking, and coding skills in Python and C++
- Collaborated and discussed solutions with peers to improve efficiency and coding techniques

ADDITIONAL

Technical Skills:

- Full Stack Development: JavaScript/TypeScript, React, Tailwind CSS, HTML/CSS, Python, FastAPI, PostgreSQL, SQLite, Docker, Git, GitHub AWS EC2
- Data Science & Machine Learning: Python, R, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Scikit-Learn Libraries
- Deep Learning: TensorFlow/Keras, PyTorch, HuggingFace Transformers, CNNs, RNNs/LSTMs, Transformers, model training and fine-tuning, GPU-accelerated computing
- Document Understanding: Multimodal LLMs, Retrieval-Augmented Generation, Agentic/Adaptive-Trajectory Pipelines, Evidence Attribution & Error Analysis, Multi-Hop Reasoning, Document Layout & OCR-based Parsing
- Research skills: Literature Review and Survey Methodology, Experimental Design, Empirical Evaluation and Error Analysis, Academic Writing, Peer Review
- Tools & Platforms: Microsoft DevOps, SharePoint, MYOB, Microsoft Office (Word, Excel, PowerPoint, Outlook)

Languages:

- English (fluent, first language)
- Mandarin (fluent)
- Japanese (basic, JLPT N5)